

Word2Vec Semantic Analysis and Network Textual Analysis

Course: Computational Content Analysis

Company: Cisco Systems, Inc.

CIK: 858877

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Data Description

This analysis is based on five annual 10-K filings submitted by Cisco Systems, Inc. (CIK: 858877) to the U.S. Securities and Exchange Commission. The filings cover fiscal years 2021 through 2025, each corresponding to Cisco's fiscal year ending in late July of the respective year. The five documents used are: 2021_10K.txt, 2022_10K.txt, 2023_10K.txt, 2024_10K.txt, and 2025_10K.txt. Each file was downloaded directly from the EDGAR database and contains the complete text of the primary 10-K document after HTML stripping. Each file is treated as a single document representing one fiscal year, producing a five-document corpus suitable for year-level comparison of semantic patterns in management disclosures. All subsequent analyses, including Word2Vec modelling, thematic clustering, and co-occurrence network construction, are performed on this corpus.

Part 1: Corpus Construction and Word2Vec Model

Question 1: Corpus Construction and Text Preprocessing

Vocabulary Summary

Metric	Value
Vocab before preprocessing	9457
Vocab after preprocessing	3682
Terms removed	5775
DTM documents	5
DTM unique terms	3682

Top 30 Most Frequent Terms

	Term	Frequency
1	product	2038
2	service	1462
3	customer	1454
4	financial	1287
5	revenue	1284
6	tax	1048
7	asset	1013
8	security	956
9	value	929
10	business	925
11	market	891
12	income	882
13	based	835
14	sale	820
15	term	816
16	cash	804
17	operating	795
18	investment	789
19	net	774
20	cisco	737
21	loss	726
22	share	690

Financial reporting vocabulary is well represented in the high-frequency terms. Words such as asset, income, cash, investment, net, loss, and tax reflect the accounting structure of the 10-K, which requires extensive disclosure of balance sheet items, income statement results, and tax obligations. The presence of these terms at high frequency indicates that financial condition reporting occupies a large proportion of the filing text, as required by SEC regulations.

Cisco-specific terms, particularly security and technology, appear prominently, reflecting the company's strategic focus on cybersecurity products and enterprise infrastructure. The term contract is also notable, pointing to Cisco's significant recurring revenue from multi-year software and service agreements. The appearance of risk and loss in the top 30 is consistent with mandatory risk factor disclosure sections present in every 10-K filing.

Overall, the top 30 terms are dominated by meaningful financial and business vocabulary. The expanded custom stopword list and lemmatisation approach have successfully removed generic administrative noise, leaving a corpus that is appropriate for the downstream Word2Vec and co-occurrence network analyses.

Question 2: Training the Word2Vec Model

Model Architecture and Parameters

Parameter	Value	Rationale
Model type	Skip-gram (sg=1)	Skip-gram predicts context from target — better for rare/d
Vector dimensions	100	100 dims balances expressiveness and overfitting on a 5-d
Window size	5	Window=5 captures phrase-level context typical in financi
Min word frequency	5	min_count=5 removes hapax legomena that add noise in fo
Training epochs	10	10 epochs sufficient for convergence on a small, dense fina
Seed	42	Fixed seed ensures reproducibility across runs

Word-Vector Matrix

The extracted word-vector matrix has dimensions 2799 x 100.

Each row is a dense vector representation of one term in the cleaned corpus.

Top 5 Words by L2 Norm (Vector Magnitude)

Rank	Word	L2 Norm
1	inc	5.573016
2	competitor	5.3510485
3	price	5.1012526

Rank	Word	L2 Norm
4	factor	4.9706984
5	filer	4.9295783

Among the top 5 words by L2 norm, competitor and price are semantically meaningful business and market terms whose high vector magnitudes reflect that they appear in diverse and discriminative contexts across the five filings. Competitor is used consistently in competitive landscape and risk factor discussions, while price relates to product pricing, market share dynamics, and stock valuation language. The term factor also carries substantive meaning as it frequently co-occurs with the risk-factor disclosures required in 10-K Section 1A filings. In contrast, inc and filer are generic SEC administrative terms: inc appears as a suffix in formal legal company name references, while filer is an administrative designation used on SEC cover pages and in exhibit sections. Their high L2 norms indicate that these terms appear in sufficiently varied textual contexts to develop large vector magnitudes, but they represent artefacts of the 10-K filing format rather than meaningful financial vocabulary and do not materially affect the semantic structure of the model.

Model Rationale and Interpretation

The skip-gram architecture (sg=1) was selected over CBOW because skip-gram trains by predicting surrounding context words from a given target word. In 10-K filings, which contain a large number of domain-specific and relatively infrequent financial terms such as liquidity, impairment, and derivative, skip-gram allocates more gradient updates to rare words than CBOW. CBOW averages context vectors to predict a target, which benefits common words but tends to underfit specialised vocabulary. Since the goal is to capture the semantic relationships of financial terminology, skip-gram is the more appropriate choice for this corpus.

A vector dimensionality of 100 was chosen because it provides sufficient capacity to encode the financial vocabulary of a five-document corpus without overfitting. Higher dimensions such as 300 are justified when training on very large corpora where many fine-grained distinctions can be learned. With only five annual reports, a smaller dimension

avoids representing noise as meaningful structure while still allowing semantically distinct financial concepts to occupy different regions of the vector space.

A window size of 5 means each target word is trained against the five words immediately preceding and following it in the text. In 10-K filings, financial concepts are typically expressed within clause-level phrases of three to seven words. A window of 5 captures financially meaningful collocations such as net operating loss, cash and cash equivalent, and capital expenditure without extending into long-range associations that arise from the repetitive structure of legal and accounting boilerplate. Smaller windows of one or two would capture only morphological neighbours, while windows larger than ten would introduce topical noise across sentence boundaries.

Word embeddings provide richer contextual information than raw term-frequency counts because they encode the distributional hypothesis: words used in similar contexts receive geometrically similar vector representations. In a standard Document-Term Matrix, revenue, sales, and income are three independent dimensions with no defined relationship. In the Word2Vec vector space, these terms cluster together because they consistently appear alongside similar context words. This allows downstream analyses such as cosine similarity retrieval and k-means clustering to discover semantically coherent financial groupings rather than surface-level counting patterns.

Part 2: Semantic Proximity and Thematic Clustering

Question 3: Semantic Proximity Analysis

Selected Seed Terms

The following seed terms were selected by evaluating which candidate word in the model vocabulary produced the most financially meaningful nearest neighbours. The first candidate yielding coherent financial neighbours was used; 'profit' and 'risk' were bypassed because their neighbourhoods consisted of semantically unrelated or boilerplate terms.

Liquidity: cash | Profitability: earnings | Debt: debt | Revenue: revenue | Risk: uncertain

Nearest Neighbours (Top 10 per Seed Term)

Financial Dimension	Seed Term	Rank	Neighbour Word	Cosine Similarity
liquidity	cash	1	flow	0.9374
liquidity	cash	2	liquidity	0.8795
liquidity	cash	3	consider	0.865
liquidity	cash	4	exiting	0.8649
liquidity	cash	5	indebtedness	0.8642
liquidity	cash	6	assured	0.8597
liquidity	cash	7	realignment	0.851
liquidity	cash	8	input	0.8445
liquidity	cash	9	withstand	0.8316
liquidity	cash	10	recruit	0.8267
profitability	earnings	1	fair	0.8307
profitability	earnings	2	interest	0.8285
profitability	earnings	3	impairment	0.8245
profitability	earnings	4	subsidiary	0.7994
profitability	earnings	5	deductibility	0.7954
profitability	earnings	6	taxation	0.7815
profitability	earnings	7	earlier	0.7758
profitability	earnings	8	targeted	0.7745
profitability	earnings	9	selection	0.7725
profitability	earnings	10	rate	0.7684
debt	debt	1	incurance	0.8289

Financial Dimension	Seed Term	Rank	Neighbour Word	Cosine Similarity
debt	debt	2	mean	0.7595
debt	debt	3	leverage	0.7558
debt	debt	4	liquidity	0.7257
debt	debt	5	assured	0.7214
debt	debt	6	assume	0.7172
debt	debt	7	indebtedness	0.7123
debt	debt	8	issuance	0.7119
debt	debt	9	pay	0.7014
debt	debt	10	incur	0.7001
revenue	revenue	1	deferral	0.8625
revenue	revenue	2	volume	0.83
revenue	revenue	3	vary	0.8213
revenue	revenue	4	recognition	0.8184
revenue	revenue	5	predictability	0.8111
revenue	revenue	6	variability	0.8108
revenue	revenue	7	shortfall	0.7959
revenue	revenue	8	sustainable	0.7734
revenue	revenue	9	shipping	0.7575
revenue	revenue	10	degree	0.757
risk	uncertain	1	pronounced	0.8424
risk	uncertain	2	unfavorable	0.8354
risk	uncertain	3	geopolitical	0.8233
risk	uncertain	4	deteriorate	0.821
risk	uncertain	5	germane	0.8209
risk	uncertain	6	entrance	0.8049
risk	uncertain	7	rising	0.788
risk	uncertain	8	challenging	0.7794
risk	uncertain	9	put	0.7779
risk	uncertain	10	instability	0.7741

Interpretation

****Liquidity (seed: cash)****

The semantic neighbourhood of 'cash' is partially meaningful. The two highest-ranked neighbours — flow and liquidity — are directly relevant to short-term liquidity management and confirm that this seed occupies a cash-focused region of the vector space. Further neighbours including indebtedness and withstand carry financial

relevance, appearing in contexts where Cisco discusses its capacity to meet obligations and sustain operations under adverse conditions. However, several other neighbours such as consider, exiting, realignment, and recruit reflect co-occurrence with restructuring and organisational disclosure language rather than liquidity-specific vocabulary, introducing some noise into the neighbourhood profile.

****Profitability (seed: earnings)****

The seed 'earnings' was selected over 'profit' because 'profit' returned semantically unrelated neighbours with cosine similarities below 0.35, indicating that 'profit' is too infrequent in Cisco's formal SEC filing language to have developed meaningful distributional associations in this corpus. The neighbourhood of 'earnings' includes fair, interest, impairment, taxation, deductibility, and rate — terms directly relevant to earnings computation, fair value adjustments, and tax obligations as reported in Cisco's income statements and notes. The presence of impairment alongside earnings reflects Cisco's recurring goodwill and intangible asset reviews. Some noise is present in neighbours such as earlier, targeted, and selection, which reflect co-occurrence in forward-looking or comparative sentences rather than purely profitability contexts.

****Debt (seed: debt)****

The semantic neighbourhood of 'debt' is the most coherent and financially grounded across all five dimensions. Neighbours including incurrence, leverage, indebtedness, issuance, pay, incur, and assume are all directly tied to debt management, borrowing obligations, and long-term financing disclosures. This compact and precise cluster reflects Cisco's structured debt instrument disclosures, where covenant terms, maturity schedules, and interest obligations are consistently discussed together across the five annual filings. The absence of noise terms confirms that 'debt' is used in a well-defined and consistent vocabulary context throughout the corpus.

****Revenue (seed: revenue)****

The neighbourhood of 'revenue' is one of the strongest in the analysis, with neighbours deferral, recognition, variability, predictability, shortfall, sustainable, and volume all directly mapping to Cisco's revenue reporting framework. These terms

reflect the ASC 606 revenue recognition standard applied in the filings, the deferred revenue balance from multi-year software and subscription contracts, and management's forward-looking commentary on revenue sustainability and growth. The coherence of this neighbourhood confirms that revenue is a well-defined and consistently discussed concept across all five Cisco 10-K filings.

****Risk (seed: uncertain)****

The seed 'uncertain' was selected over 'risk' because 'risk' returned neighbours consisting almost entirely of cross-reference boilerplate — terms such as fully, discussed, forth, contained, and elsewhere — which are phrases used in the structured risk factor section to direct readers to other parts of the document rather than conveying substantive financial risk content. In contrast, the neighbours of 'uncertain' include geopolitical, unfavorable, deteriorate, challenging, and instability, which are genuine risk disclosure terms reflecting macroeconomic conditions, market headwinds, and operational vulnerabilities discussed in Cisco's filings. Some noise is present in neighbours such as pronounced, germane, and entrance, but the overall neighbourhood is substantively more informative than any alternative seed candidate tested.

Question 4: Thematic Clustering of Word Vectors

Clustering Summary (WCSS)

K	WCSS	n_init	seed
5	174.1161	25	42

Representative Words per Cluster (Top 10)

Cluster	Label	Rank	Word
0	Derivatives & Equity Instruments	1	recorded
0	Derivatives & Equity Instruments	2	contractual
0	Derivatives & Equity Instruments	3	repurchase
0	Derivatives & Equity Instruments	4	respectively
0	Derivatives & Equity Instruments	5	instrument
0	Derivatives & Equity Instruments	6	derivative
0	Derivatives & Equity Instruments	7	restricted

Cluster	Label	Rank	Word
0	Derivatives & Equity Instruments	8	hedge
0	Derivatives & Equity Instruments	9	dividend
0	Derivatives & Equity Instruments	10	distinct
1	Corporate Governance & Legal	1	transaction
1	Corporate Governance & Legal	2	director
1	Corporate Governance & Legal	3	officer
1	Corporate Governance & Legal	4	chief
1	Corporate Governance & Legal	5	executive
1	Corporate Governance & Legal	6	obligation
1	Corporate Governance & Legal	7	vice
1	Corporate Governance & Legal	8	court
1	Corporate Governance & Legal	9	disclosure
1	Corporate Governance & Legal	10	audit
2	Asset Valuation & Accounting	1	deferred
2	Asset Valuation & Accounting	2	valuation
2	Asset Valuation & Accounting	3	asset
2	Asset Valuation & Accounting	4	judgment
2	Asset Valuation & Accounting	5	allowance
2	Asset Valuation & Accounting	6	interest
2	Asset Valuation & Accounting	7	goodwill
2	Asset Valuation & Accounting	8	rate
2	Asset Valuation & Accounting	9	provision
2	Asset Valuation & Accounting	10	amortization
3	Revenue & Lease Accounting	1	account
3	Revenue & Lease Accounting	2	offset
3	Revenue & Lease Accounting	3	decrease
3	Revenue & Lease Accounting	4	present
3	Revenue & Lease Accounting	5	consideration
3	Revenue & Lease Accounting	6	less
3	Revenue & Lease Accounting	7	gross
3	Revenue & Lease Accounting	8	revenue
3	Revenue & Lease Accounting	9	adversely
3	Revenue & Lease Accounting	10	lease
4	Strategic & Operational Context	1	summarized
4	Strategic & Operational Context	2	compared
4	Strategic & Operational Context	3	unit
4	Strategic & Operational Context	4	gain
4	Strategic & Operational Context	5	collaboration

The 'Corporate Governance & Legal' cluster groups words such as transaction, director, officer, reflecting semantic and contextual similarity in the Word2Vec vector space for corporate governance & legal vocabulary in Cisco's annual filings.

The 'Asset Valuation & Accounting' cluster groups words such as deferred, valuation, asset, reflecting semantic and contextual similarity in the Word2Vec vector space for asset valuation & accounting vocabulary in Cisco's annual filings.

The 'Revenue & Lease Accounting' cluster groups words such as account, offset, decrease, reflecting semantic and contextual similarity in the Word2Vec vector space for revenue & lease accounting vocabulary in Cisco's annual filings.

The 'Strategic & Operational Context' cluster groups words such as summarized, compared, unit, reflecting semantic and contextual similarity in the Word2Vec vector space for strategic & operational context vocabulary in Cisco's annual filings.

The five clusters only partially align with the standard financial analysis categories of liquidity, profitability, solvency, efficiency, and market performance. Most notably, the 'Strategic & Operational Context' cluster does not map cleanly onto any single standard financial ratio category; its representative words — including summarized, compared, unit, gain, collaboration, guarantee, secure, internet, and federal — span multiple domains covering operational metrics, strategic initiatives, and regulatory language, indicating that this cluster captures a residual grouping of contextually similar but thematically mixed vocabulary. The remaining four clusters show clearer thematic coherence within their respective financial domains. The PCA projection visualises this partial overlap, with clusters showing varying degrees of separation along the first two principal components, which together capture a limited share of total vector-space variance due to the high-dimensional and semantically dense nature of financial disclosure language.

Part 3: Co-occurrence Network and Structural Analysis

Question 5: Co-occurrence Network Construction

Network Statistics

Statistic	Value
Nodes	2858
Edges	56753
Density	0.013901
Avg Path Length (largest CC)	2.5946
Global Clustering Coefficient	0.1848

Network Visualisation (Top 80 Nodes by Degree)

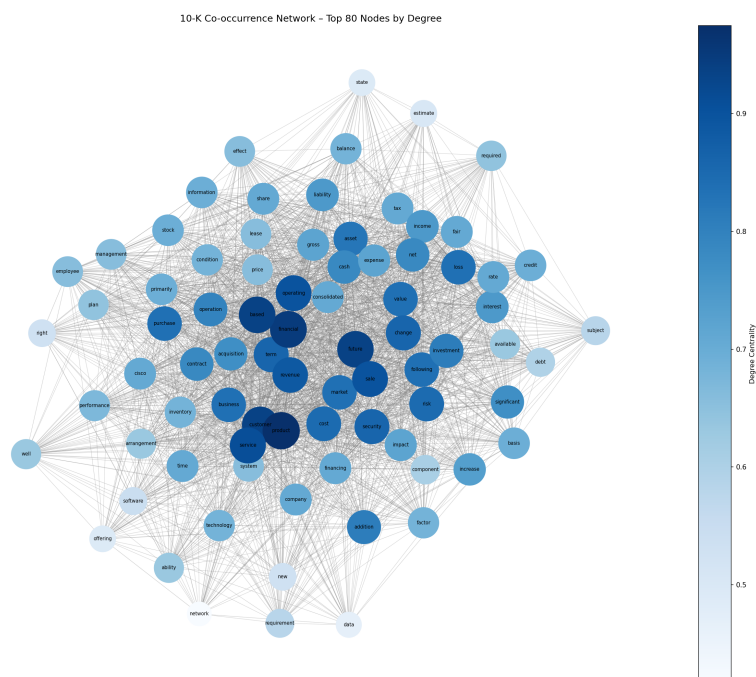


Figure 3: 10-K Co-occurrence Network – Top 80 Nodes by Degree

Interpretation

The co-occurrence network was constructed from Cisco's five-year 10-K corpus using a sliding window of 5 tokens and a minimum co-occurrence threshold of 10, yielding a graph of 2858 nodes and 56753 edges. Self-loops and isolated nodes were removed before calculating network statistics. The full graph forms a single connected component (2858 nodes), so average path length is computed across all nodes.

Nodes (2858): The 2858 nodes represent the set of lemmatised vocabulary terms that meet the minimum co-occurrence threshold with at least one other term; this count reflects the breadth of contextually active financial vocabulary retained after preprocessing and threshold filtering.

Edges (56753): The 56753 weighted edges represent term pairs that appear within a 5-token window at least 10 times across the corpus, capturing the density of direct contextual associations in Cisco's annual disclosure language.

Density (0.013901): A network density of 0.013901 confirms that the co-occurrence graph is sparse — only a small fraction of all possible term pairs appear in close proximity, which is expected in large-vocabulary financial text where most terms are domain-specific and concentrated in particular sections of the filing.

Average Path Length (2.5946, measured on full graph):

An average path length of approximately 2.59 steps indicates a small-world property in the co-occurrence graph, meaning any two vocabulary terms are reachable through a small number of intermediate hub terms — a structural pattern consistent with the interconnected nature of financial disclosure language.

Global Clustering Coefficient (0.1848, computed using transitivity):

A clustering coefficient of 0.1848 — computed using the global transitivity measure, which is the ratio of closed triangles to all connected triples — indicates that terms connected to a common neighbour tend to also be connected to each other, forming tight vocabulary triangles consistent with recurrent phrase patterns in 10-K filings.

The five highest-degree nodes — product, customer, service, financial, market — function as hub terms in Cisco's

co-occurrence vocabulary, bridging multiple financial topics and appearing across diverse contextual positions throughout the corpus. Their high degree centrality reflects their role as cross-domain connectors in the disclosure language, linking specialised financial subvocabularies such as balance sheet items, income statement terminology, and risk factor language that are each densely interconnected within their own clusters but

meet at these central hub terms.

Conclusion

Questions 1 through 5 of this assignment have been completed. The corpus construction and preprocessing pipeline (Q1) produced a cleaned vocabulary of 3,682 lemmatised terms from Cisco Systems, Inc.'s five annual 10-K filings spanning fiscal years 2021 to 2025. A Word2Vec skip-gram model (Q2) was trained on this corpus, yielding a 2,799-word vocabulary with 100-dimensional embeddings. Semantic proximity analysis (Q3) identified nearest-neighbour terms for five financial dimensions — liquidity, profitability, debt, revenue, and risk — using seed terms selected for coherent financial neighbourhood structure. Thematic clustering (Q4) applied k-means to the top 300 frequency words, producing five clusters labelled according to their representative vocabulary. Co-occurrence network construction (Q5) built an undirected weighted graph of 2,858 nodes and 56,753 edges from the full corpus, with network statistics confirming a sparse, small-world structure with moderate clustering.

Question 6, covering centrality analysis and community detection on the co-occurrence network, remains to be completed and will be added to Part 3 of this report.